Hints: Combinatorics – 2. Recurrences and generating functions

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $p(n-1)$ counts the partitions of n that contain at least one part 1: remove one 1. The rest of the partitions of n have all parts $\geqslant 2$. (In generating functions: multiply $\prod_{k\geqslant 1}(1-x^k)^{-1}$ by $1-x$.)

Problem 2

Hint 1. By the formal binomial series, $[x^n](1-4x)^{-1/2} = \binom{-1/2}{n}(-4)^n$. Write $\binom{-1/2}{n} = \frac{(-1)^n\,1\cdot 3\cdots(2n-1)}{2^n\,n!}$ and multiply out.

Hint 2. $\sum_k \binom{n}{k}^2 = \sum_k \binom{n}{k}\binom{n}{n-k}$ is the coefficient of x^n in $(1+x)^n(1+x)^n = (1+x)^{2n}$. Double count: choose n people out of n men and n women, classifying by the number k of men. (Squaring the series also gives $\sum_k \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n$.)

Problem 3

Hint 1. Every permutation is a set of fixed points plus a derangement of the rest: $n! = \sum_k \binom{n}{k} D_{n-k}$, i.e. $\frac{1}{1-x} = e^x D(x)$ for exponential generating functions.

Hint 2. An involution is a set of cycles of length 1 or 2; the EGF of a single such component is $x + x^2/2$, so the involutions have $e^{x+x^2/2}$. A set partition is a set of non-empty blocks, a block has EGF $e^x - 1$, so Bell numbers have e^{e^x-1}.

Hint 3. Differentiate $I(x) = e^{x+x^2/2}$: $I' = (1+x)I$, i.e. $I_{n+1} = I_n + nI_{n-1}$ (combinatorially: $n+1$ is fixed or paired with one of the other n).

Problem 4

Hint 1. For 101, the states are the longest suffix that is a proper prefix of 101: ε (or ending in 00), 1, 10. Write the 3×3 transfer matrix.

Hint 2. Avoiding 101 gives $a_n = 2a_{n-1} - a_{n-2} + a_{n-3}$ $(2, 4, 7, 12, 21, 37, \ldots)$, characteristic polynomial $t^3 - 2t^2 + t - 1$, growth rate ≈ 1.755. Avoiding 11 gives Fibonacci growth $\frac{1+\sqrt{5}}{2} \approx 1.618$.

Problem 5

Hint 1. Let there be n digits, z of them zeros; by symmetry assume $z \leqslant n/2$. Show that two zeros are never adjacent, and more generally that the gaps of ones between consecutive zeros take only two consecutive values $m, m+1$.

Hint 2. Replace each block "$0\,1^m$" by one letter and each "$0\,1^{m+1}$" by another; show the new circular word is again balanced (an arc of the new word corresponds to an arc of the old one). This is a Euclidean-algorithm descent on (n, z).

Hint 3. Conclude by induction that the balanced collections are exactly the rotations of the "mechanical" word in which position k $(1 \leqslant k \leqslant n)$ is a zero iff $\lfloor kz/n \rfloor > \lfloor (k-1)z/n \rfloor$, i.e. zeros

as evenly spread as possible; conversely check this word is balanced, since every arc of length L contains $\lfloor Lz/n \rfloor$ or $\lceil Lz/n \rceil$ zeros.

Problem 6

Hint 1. Divide the relation by $a_{n-1}a_{n-2}a_{n-3}$ and look at the ratios $b_n = a_n/a_{n-1}$.

Hint 2. $b_n = b_{n-1} + b_{n-2}$ with $b_2 = 1$, $b_3 = 2$, so $b_n = F_n$ and $a_n = F_1F_2 \cdots F_n$.

Hint 3. Imitate the proof that e is irrational. If $S = p/q$, pick N with $q \mid a_N$ (every integer divides some Fibonacci number). Then $a_N S - \sum_{n \leqslant N} a_N/a_n = \sum_{n>N} a_N/a_n$ is a positive integer, but it is less than $\frac{1}{F_{N+1}} + \frac{1}{F_{N+1}F_{N+2}} + \ldots < 1$ for large N.

Problem 7

Hint 1. For a path ending at (a, b) with $a \geqslant b$ that touches $y = x + 1$, reflect the part of the path before the first touching point in the line $y = x + 1$. Where does the starting point $(0, 0)$ go?

Hint 2. The reflected path runs from $(-1, 1)$ to (a, b); this is a bijection with all paths between these points, i.e. (after translation) with paths to $(a + 1, b - 1)$: there are $\binom{a+b}{a+1}$ of them. Hence paths staying weakly below the diagonal number $\binom{a+b}{a} - \binom{a+b}{a+1} = \frac{a-b+1}{a+1}\binom{a+b}{a}$ (ballot), and $a = b = n$ gives $C_n = \frac{1}{n+1}\binom{2n}{n}$.

Problem 8

Hint 1. Fix n and find $\sum_m f(m, n)x^m$: the generating function for one integer t by $|t|$ is $\frac{1+x}{1-x}$, and "$\leqslant m$" contributes an extra factor $\frac{1}{1-x}$.

Hint 2. So $\sum_{m,n} f(m, n)x^m y^n = \frac{1}{1-x} \sum_n \left(\frac{y(1+x)}{1-x}\right)^n = \frac{1}{1-x-y-xy}$, which is symmetric in x, y.

Problem 9

Hint 1. $2 \times n$: the leftmost column is a vertical domino or the start of two horizontal ones, so $d_n = d_{n-1} + d_{n-2}$, $d_n = F_{n+1}$. For $3 \times 2n$, look at the first place where a vertical cut does not cross any domino.

Hint 2. The first such cut is at width 2 (3 ways) or at an even width $2j \geqslant 4$ (2 ways each), so $t_n = 3t_{n-1} + 2(t_{n-2} + \cdots + t_0)$; subtract the same identity for $n - 1$ to get $t_n = 4t_{n-1} - t_{n-2}$, $t_0 = 1$, $t_1 = 3$.

Hint 3. Generating function $\frac{1-x}{1-4x+x^2}$, closed form $t_n = \frac{(3+\sqrt{3})(2+\sqrt{3})^n + (3-\sqrt{3})(2-\sqrt{3})^n}{6}$.

Problem 10

Hint 1. ($\Rightarrow$) If $a_n = \sum_{i=1}^k c_i a_{n-i}$ for $n \geqslant n_0$, then in $H_N = (a_{i+j})_{0 \leqslant i,j \leqslant N}$ with $N \geqslant n_0$ the row with index n_0 is a linear combination of the k rows above it.

Hint 2. ($\Leftarrow$) Let $D_N = \det H_N$ and take p minimal with $D_N = 0$ for all $N \geqslant p$ (with the convention $D_{-1} = 1$); then $D_{p-1} \neq 0$, so the last row of H_p is a combination of the first p: $a_{p+j} = \sum_{i<p} c_i a_{i+j}$ for $0 \leqslant j \leqslant p$. Let $\varepsilon_j = a_{p+j} - \sum c_i a_{i+j}$ and suppose m is the first index with $\varepsilon_m \neq 0$.

Hint 3. In H_m subtract from each row $r \geqslant p$ the combination of rows $r - p, \ldots, r - 1$: rows $p, \ldots, m$ become $(\varepsilon_{r-p+j})_j$, which vanish for $r - p + j < m$. The matrix becomes block triangular with an anti-triangular block having ε_m on the antidiagonal, so $D_m = \pm D_{p-1}\varepsilon_m^{m-p+1} \neq 0$, a contradiction. For $\binom{2n}{n}$: either compute the Hankel determinants (they equal 2^N), or note that $(1 - 4x)^{-1/2}$ is not rational because $1 - 4x$ is not a square of a rational function; so no recurrence.